



Cambridge IGCSE™

CHEMISTRY

0620/21

Paper 2 Multiple Choice (Extended)

October/November 2025

45 minutes

You must answer on the multiple choice answer sheet.

You will need: Multiple choice answer sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)

INSTRUCTIONS

- There are **forty** questions on this paper. Answer **all** questions.
- For each question there are four possible answers **A**, **B**, **C** and **D**. Choose the **one** you consider correct and record your choice in soft pencil on the multiple choice answer sheet.
- Follow the instructions on the multiple choice answer sheet.
- Write in soft pencil.
- Write your name, centre number and candidate number on the multiple choice answer sheet in the spaces provided unless this has been done for you.
- Do **not** use correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.

INFORMATION

- The total mark for this paper is 40.
- Each correct answer will score one mark.
- Any rough working should be done on this question paper.
- The Periodic Table is printed in the question paper.

This document has **16** pages. Any blank pages are indicated.



- 1 A sealed balloon contains air.

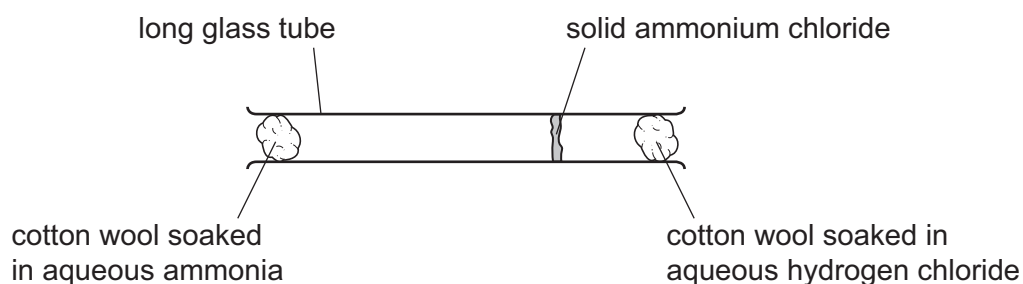
The temperature of the balloon is increased.

Which row describes how the pressure in the balloon and the kinetic energy of the particles in air change?

	pressure	kinetic energy of particles
A	decreases	decreases
B	increases	decreases
C	decreases	increases
D	increases	increases

- 2 Ammonia gas is reacted with hydrogen chloride gas using the apparatus shown.

Solid ammonium chloride is produced.



Which statement explains why the solid ammonium chloride is formed nearer to the hydrogen chloride?

- A** Aqueous ammonia is a base and aqueous hydrogen chloride is an acid.
- B** Ammonia molecules diffuse more slowly than hydrogen chloride molecules.
- C** Hydrogen chloride has a greater molecular mass than ammonia.
- D** Only ammonia molecules diffuse in all directions at the same time.

- 3 The table shows information about four different particles.

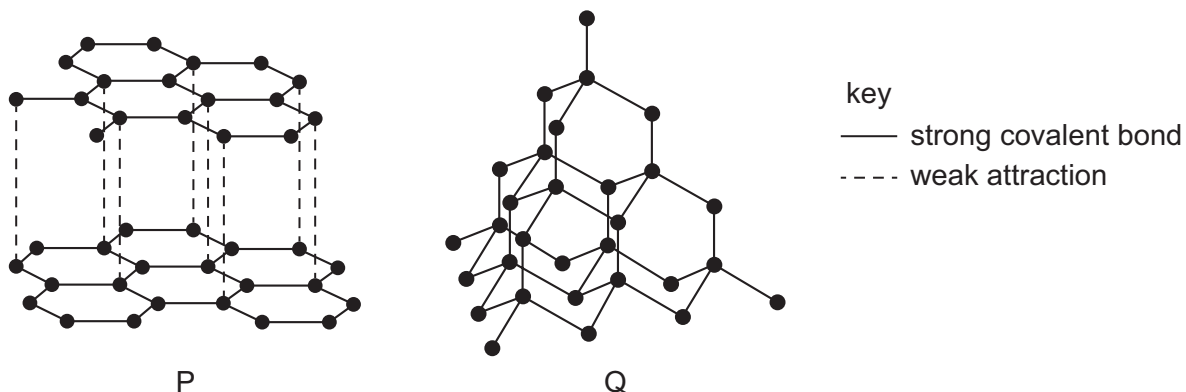
particle	nucleon number	number of protons	number of neutrons	number of electrons
Na	23	11	W	11
Na ⁺	23	11	12	X
O	16	8	Y	8
O ²⁻	16	8	8	Z

What are the values of W, X, Y and Z?

	W	X	Y	Z
A	11	10	10	8
B	11	11	8	10
C	12	10	8	10
D	12	11	10	8

- 4 Substances with giant covalent structures can be used as lubricants and as cutting tools for hard materials.

The diagrams show how the atoms are arranged in two giant covalent substances, P and Q.



Which statement is correct?

- A** Only P is used as a cutting tool and only Q is used as a lubricant.
- B** Only P is used as a lubricant and only Q is used as a cutting tool.
- C** P and Q are both used as cutting tools.
- D** P and Q are both used as lubricants.

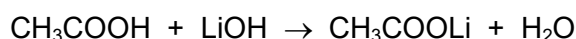
- 5 Symbols representing four particles are shown.



The letters are **not** the chemical symbols.

Which particles have the same number of neutrons?

- A** W and X^{2+} **B** W and Z **C** X^{2+} and Y **D** Y and Z
- 6 Which statement describes the formation of bonds in lithium bromide?
- A** A lithium atom donates one electron to a bromine atom.
B A bromine atom donates one electron to a lithium atom.
C A lithium atom shares a pair of electrons with a bromine atom.
D A bromine atom donates a pair of electrons to a lithium atom.
- 7 Which statement explains why solid magnesium is a good conductor of electricity?
- A** Magnesium has delocalised electrons that can move through the lattice.
B Magnesium has strong attractions between ions.
C Magnesium consists of a giant lattice of anions in a 'sea' of delocalised electrons.
D Magnesium consists of metal ions that move through the lattice.
- 8 The equation for the reaction between dilute ethanoic acid and aqueous lithium hydroxide is shown.



In an experiment, 20.0 cm^3 of 0.050 mol/dm^3 aqueous lithium hydroxide is exactly neutralised by 12.5 cm^3 of dilute ethanoic acid.

Which statement is correct?

- A** The amount of lithium hydroxide is equal to $0.050 \times \frac{1000}{20}$ mol.
B The amount of ethanoic acid reacted is 0.0010 mol.
C The concentration of the dilute ethanoic acid is 0.16 mol/dm^3 .
D The volume of aqueous lithium hydroxide used is 0.20 dm^3 .

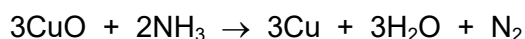
9 The molecular formulae of four compounds are listed.

- 1 CH₄
- 2 C₂H₆
- 3 C₃H₈
- 4 C₃H₆

Which molecular formulae are also the empirical formulae for the compounds?

- A** 1 and 2 **B** 1 and 3 **C** 2 and 4 **D** 3 and 4

10 The equation for the reaction between copper(II) oxide, CuO, and ammonia is given.



[*M_r*: CuO, 80; NH₃, 17; H₂O, 18; N₂, 28]

[*A_r*: Cu, 64]

Which statements about this reaction are correct?

- 1 80.0 g of copper(II) oxide reacts exactly with 17.0 g of ammonia.
- 2 4.0 g of copper(II) oxide reacts with excess ammonia to produce 3.2 g of copper.
- 3 The mass of water produced in this reaction is greater than the mass of nitrogen.

- A** 1, 2 and 3 **B** 1 and 2 only **C** 1 and 3 only **D** 2 and 3 only

11 How many atoms are present in 1.00 mol of argon?

- A** 2.06×10^{23} **B** 3.02×10^{26} **C** 6.02×10^{23} **D** 6.32×10^{20}

12 Aqueous copper(II) sulfate is electrolysed using copper electrodes.

Which statement is correct?

- A** Oxygen gas is produced at the positive electrode.
- B** The blue colour of the solution gradually fades.
- C** The concentration of copper ions in the solution stays the same.
- D** The mass of the negative electrode decreases.

13 Which row describes the reactions during the electrolysis of dilute aqueous sodium chloride?

	anode reaction	cathode reaction
A	$\text{H}_2 \rightarrow 2\text{H}^+ + 2\text{e}^-$	$2\text{H}_2\text{O} + \text{O}_2 + 4\text{e}^- \rightarrow 4\text{OH}^-$
B	$2\text{H}^+ + 2\text{e}^- \rightarrow \text{H}_2$	$4\text{OH}^- \rightarrow 2\text{H}_2\text{O} + \text{O}_2 + 4\text{e}^-$
C	$2\text{H}_2\text{O} + \text{O}_2 + 4\text{e}^- \rightarrow 4\text{OH}^-$	$\text{H}_2 \rightarrow 2\text{H}^+ + 2\text{e}^-$
D	$4\text{OH}^- \rightarrow 2\text{H}_2\text{O} + \text{O}_2 + 4\text{e}^-$	$2\text{H}^+ + 2\text{e}^- \rightarrow \text{H}_2$

14 Information about two reactions is given.

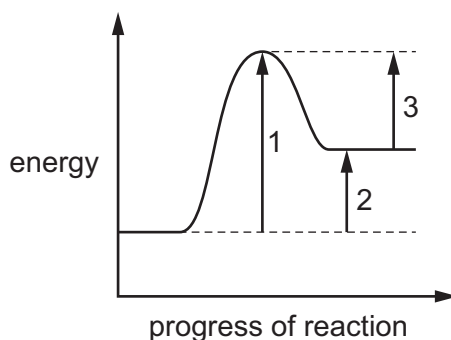
- The neutralisation reaction between citric acid and sodium hydrogencarbonate is endothermic.
- The displacement reaction between magnesium and carbon dioxide is exothermic.

Which statements about the two reactions are correct?

- 1 The energy of the products formed in the neutralisation reaction is greater than the energy of the reactants.
- 2 The energy of magnesium and carbon dioxide is greater than the energy of magnesium oxide and carbon.
- 3 In an exothermic reaction, the energy required to break the bonds is greater than the energy released when the new bonds are formed.

A 1, 2 and 3 **B** 1 and 2 only **C** 1 and 3 only **D** 2 and 3 only

15 The reaction pathway diagram for a reaction is shown.



Which row identifies the activation energy, E_a , and the enthalpy change, ΔH , for this reaction?

	E_a	ΔH
A	1	2
B	1	3
C	2	3
D	3	1

- 16** Hydrogen peroxide solution decomposes very slowly at room temperature to produce oxygen gas. This gas forms a rising foam when liquid detergent is added.

Five test-tubes are half-filled with hydrogen peroxide solution. A drop of liquid detergent is added to each one.

Different metal oxides are added to four of the test-tubes and the height of the foam formed after one minute is measured. The results are shown.

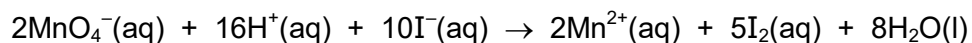
metal oxide	height of foam / cm
no metal oxide added	0.1
aluminium oxide	0.1
calcium oxide	0.2
copper(II) oxide	2.3
manganese(IV) oxide	5.4

Which conclusion can be drawn from these results?

- A** Metal oxides do **not** affect the rate of this reaction.
 - B** All metal oxides increase the rate of this reaction and act as catalysts.
 - C** Manganese(IV) oxide is the best catalyst of the four metal oxides tested.
 - D** Only transition element oxides increase the rate of this reaction.
- 17** Which equation represents one of the reactions in the Contact process?

- A** $2\text{SO}_2(\text{g}) + \text{O}_2(\text{g}) \rightleftharpoons 2\text{SO}_3(\text{g})$
- B** $2\text{SO}_2(\text{g}) + \text{O}_2(\text{g}) \rightleftharpoons 2\text{SO}_3(\text{aq})$
- C** $2\text{SO}_3(\text{g}) + \text{O}_2(\text{g}) \rightleftharpoons 2\text{SO}_4(\text{aq})$
- D** $2\text{SO}_3(\text{g}) + \text{O}_2(\text{g}) \rightleftharpoons 2\text{SO}_4(\text{g})$

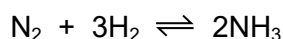
- 18 The ionic equation for the redox reaction between acidified manganate(VII) ions and iodide ions is shown.



Which row shows the changes in oxidation number for manganese and for iodine during this reaction?

	manganese		iodine	
	from	to	from	to
A	−2	+4	−1	0
B	−2	+4	−10	+5
C	+7	+2	−1	0
D	+7	+2	−10	+5

- 19 In the Haber process, nitrogen and hydrogen react to form ammonia.



The forward reaction is exothermic.

Which changes in pressure and temperature increase the yield of ammonia?

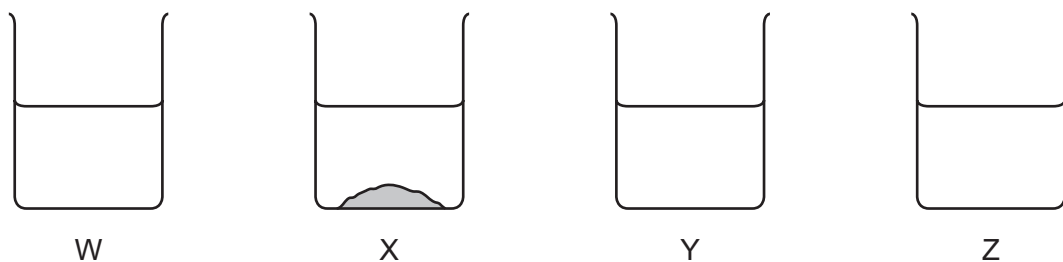
	change in pressure	change in temperature
A	decrease	decrease
B	decrease	increase
C	increase	decrease
D	increase	increase

- 20 Which statement describes a property of ethanoic acid?

- A** It does **not** produce carbon dioxide when added to sodium carbonate.
- B** It does **not** produce hydrogen when added to magnesium.
- C** It only donates some of its protons when added to excess aqueous sodium hydroxide.
- D** It only partially dissociates when added to water.

- 21** A student adds different solid salts to an excess of water. The student stirs the mixtures to determine the solubilities of the salts.

The results are shown.



What could the four different salts be?

	W	X	Y	Z
A	calcium sulfate	lead(II) chloride	ammonium nitrate	potassium sulfate
B	ammonium nitrate	sodium carbonate	calcium sulfate	lead(II) chloride
C	sodium carbonate	calcium sulfate	potassium sulfate	ammonium nitrate
D	lead(II) chloride	ammonium nitrate	sodium carbonate	calcium sulfate

- 22** Four metal oxides are listed.

- 1 Al_2O_3
- 2 CaO
- 3 CuO
- 4 ZnO

Which metal oxides are amphoteric?

- A** 1 and 4 **B** 1 only **C** 2, 3 and 4 **D** 2 and 3 only

23 The ions formed when element X and element Y react together are shown.

- Element X forms X^- ions.
- Element Y forms Y^{2+} ions.

Which row identifies the position of X and of Y in the Periodic Table?

	X	Y
A	Group I	Group VI
B	Group VII	Group II
C	Group VII	Group VI
D	Group I	Group II

24 Which properties decrease down the group of alkali metals?

- 1 melting point
- 2 density
- 3 reactivity

A 1 and 2 **B** 1 only **C** 2 and 3 **D** 3 only

25 Some halide ions are displaced from aqueous potassium halides by halogens.

In which mixtures are the halide ions displaced?

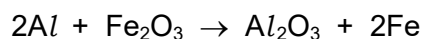
- 1 potassium bromide and chlorine
- 2 potassium bromide and iodine
- 3 potassium iodide and chlorine
- 4 potassium chloride and iodine

A 1 and 3 **B** 1 and 4 **C** 2 and 3 **D** 2 and 4

26 What is the correct order of reactivity of the following metals, from least reactive to most reactive?

- A** calcium → magnesium → zinc → copper
- B** copper → zinc → magnesium → calcium
- C** copper → zinc → calcium → magnesium
- D** zinc → copper → magnesium → calcium

- 27 The equation for the reaction between aluminium and iron(III) oxide is shown.



Which statement about this reaction is correct?

- A The aluminium acts as an oxidising agent.
- B It shows that iron is more reactive than aluminium.
- C The oxidation number of the aluminium decreases during the reaction.
- D It shows that aluminium has a greater tendency to form ions than iron.

- 28 Which row identifies methods used to prevent iron from rusting?

	greasing	painting	washing with distilled water
A	✓	✗	✓
B	✗	✓	✓
C	✓	✓	✗
D	✓	✓	✓

key

✓ = yes

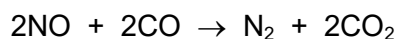
✗ = no

- 29 Which statement about the extraction of aluminium by electrolysis is correct?

- A Aluminium oxide is dissolved in molten cryolite to increase its melting point.
- B At the cathode, oxygen ions lose electrons to form oxygen gas.
- C The carbon in the graphite anode reacts with oxygen gas forming carbon dioxide.
- D The reaction at the anode is $Al^{3+} + 3e^- \rightarrow Al$.

- 30 Catalytic converters are used to remove oxides of nitrogen from exhaust gases.

The equation represents one of the reactions that occur in a catalytic converter.



What is a disadvantage of using a catalytic converter to remove oxides of nitrogen?

- A CO is an essential reactant in photosynthesis.
- B CO₂ is a greenhouse gas.
- C N₂ is acidic.
- D N₂ is toxic.

- 31 Some gases are present in clean, dry air, while other gases are **only** present in polluted air.

Which row is correct?

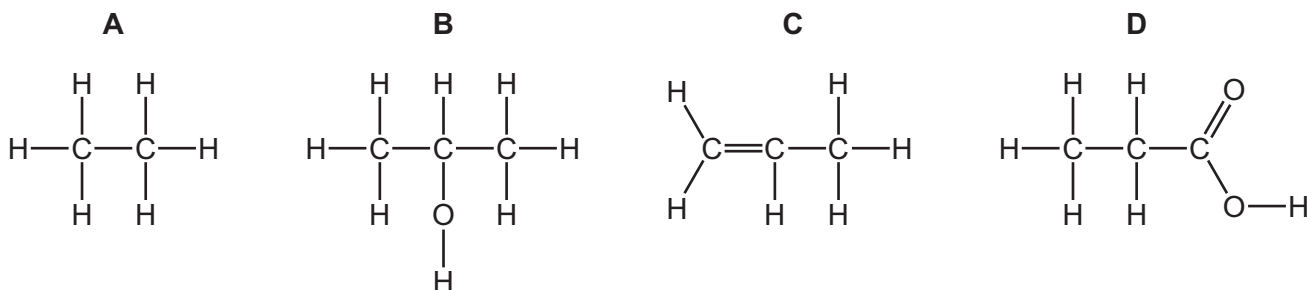
	a gas present in clean, dry air	a gas only present in polluted air
A	argon	carbon dioxide
B	argon	nitrogen dioxide
C	sulfur dioxide	carbon dioxide
D	sulfur dioxide	nitrogen dioxide

- 32 Petroleum is separated by fractional distillation.

Which statement about the fractions produced is correct?

- A** Bottled gas for heating and cooking is obtained from the naphtha fraction.
B Diesel oil is used as a fuel for jet aircraft.
C Substances used to make polishes are obtained from the lubricating fraction.
D The kerosene fraction contains many useful waxes.

- 33 Which structure represents a compound in the alcohol homologous series?

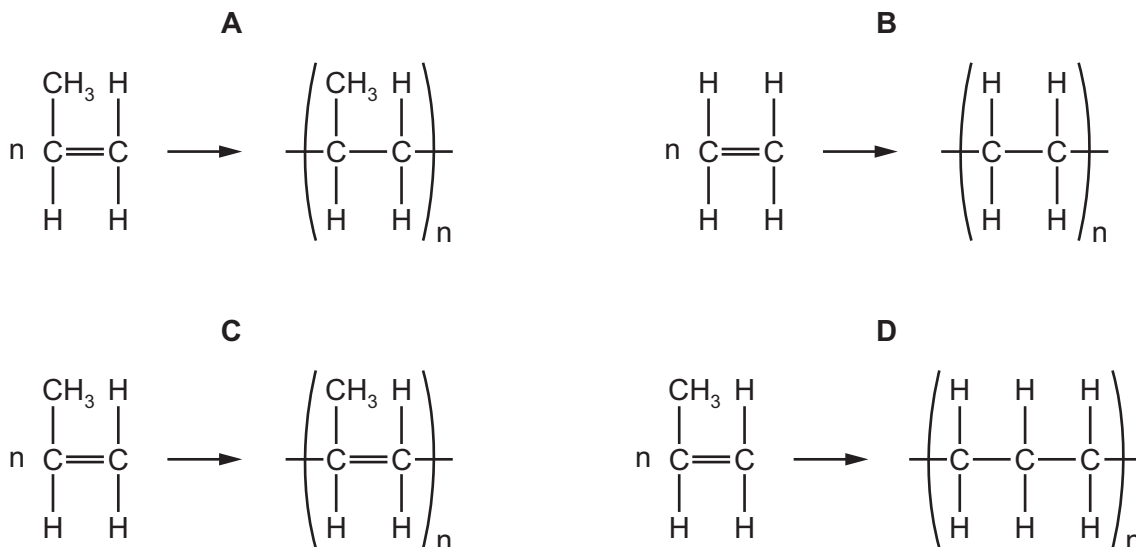


- 34 Ethane reacts with chlorine.

Which row identifies the type of reaction and the formulae of the products formed?

	reaction	products
A	addition	$\text{CH}_3\text{CH}_2\text{Cl} + \text{H}_2$
B	addition	$\text{CH}_3\text{CH}_2\text{Cl} + \text{HCl}$
C	substitution	$\text{CH}_3\text{CH}_2\text{Cl} + \text{HCl}$
D	substitution	$\text{CH}_3\text{CH}_2\text{Cl} + \text{H}_2$

35 Which equation represents the formation of poly(propene) from propene?



36 Ethene reacts with hydrogen in an addition reaction to form ethane.

Which catalyst is used for this reaction?

- A** iron
- B** nickel
- C** vanadium(V) oxide
- D** yeast

37 What is produced by the bacterial oxidation of ethanol?

- A** ethene
- B** sodium ethanoate
- C** ethyl ethanoate
- D** vinegar

38 Which statement about addition polymers or condensation polymers is correct?

- A** Addition polymers only contain carbon and hydrogen atoms.
- B** One of the monomers used to make a polyester is a diamine.
- C** Addition polymers contain carbon–carbon double bonds in the main chain.
- D** Water is an additional product in the formation of a protein from amino acids.

39 The results of flame tests on three metal salts, X, Y and Z, are shown.

metal salt	flame test colour
X	yellow
Y	lilac
Z	blue-green

Which row identifies the cations present in X, Y and Z?

	X	Y	Z
A	potassium	sodium	calcium
B	potassium	sodium	copper
C	sodium	potassium	calcium
D	sodium	potassium	copper

40 Ethanol can be made by fermentation of sugar, using yeast.

This produces a mixture of ethanol and water.

How is ethanol separated from this mixture?

- A** distilling the mixture using a fractionating column
- B** filtering the mixture
- C** heating to evaporate most of the water, and allowing the ethanol to crystallise
- D** heating to evaporate the water

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The Periodic Table of Elements

Group																		
I	II											III	IV	V	VI	VII	VIII	
3 Li lithium 7	4 Be beryllium 9	Key atomic number atomic symbol name relative atomic mass										1 H hydrogen 1	5 B boron 11	6 C carbon 12	7 N nitrogen 14	8 O oxygen 16	9 F fluorine 19	10 Ne neon 20
11 Na sodium 23	12 Mg magnesium 24											13 Al aluminium 27	14 Si silicon 28	15 P phosphorus 31	16 S sulfur 32	17 Cl chlorine 35.5	18 Ar argon 40	
19 K potassium 39	20 Ca calcium 40	21 Sc scandium 45	22 Ti titanium 48	23 V vanadium 51	24 Cr chromium 52	25 Mn manganese 55	26 Fe iron 56	27 Co cobalt 59	28 Ni nickel 59	29 Cu copper 64	30 Zn zinc 65	31 Ga gallium 70	32 Ge germanium 73	33 As arsenic 75	34 Se selenium 79	35 Br bromine 80	36 Kr krypton 84	
37 Rb rubidium 85	38 Sr strontium 88	39 Y yttrium 89	40 Zr zirconium 91	41 Nb niobium 93	42 Mo molybdenum 96	43 Tc technetium —	44 Ru ruthenium 101	45 Rh rhodium 103	46 Pd palladium 106	47 Ag silver 108	48 Cd cadmium 112	49 In indium 115	50 Sn tin 119	51 Sb antimony 122	52 Te tellurium 128	53 I iodine 127	54 Xe xenon 131	
55 Cs caesium 133	56 Ba barium 137	57–71 lanthanoids	72 Hf hafnium 178	73 Ta tantalum 181	74 W tungsten 184	75 Re rhenium 186	76 Os osmium 190	77 Ir iridium 192	78 Pt platinum 195	79 Au gold 197	80 Hg mercury 201	81 Tl thallium 204	82 Pb lead 207	83 Bi bismuth 209	84 Po polonium —	85 At astatine —	86 Rn radon —	
87 Fr francium —	88 Ra radium —	89–103 actinoids	104 Rf rutherfordium —	105 Db dubnium —	106 Sg seaborgium —	107 Bh bohrium —	108 Hs hassium —	109 Mt meitnerium —	110 Ds darmstadtium —	111 Rg roentgenium —	112 Cn copernicium —	113 Nh nihonium —	114 Fl flerovium —	115 Mc moscovium —	116 Lv livermorium —	117 Ts tennessine —	118 Og oganesson —	

lanthanoids	57 La lanthanum 139	58 Ce cerium 140	59 Pr praseodymium 141	60 Nd neodymium 144	61 Pm promethium —	62 Sm samarium 150	63 Eu europium 152	64 Gd gadolinium 157	65 Tb terbium 159	66 Dy dysprosium 163	67 Ho holmium 165	68 Er erbium 167	69 Tm thulium 169	70 Yb ytterbium 173	71 Lu lutetium 175
actinoids	89 Ac actinium —	90 Th thorium 232	91 Pa protactinium 231	92 U uranium 238	93 Np neptunium —	94 Pu plutonium —	95 Am americium —	96 Cm curium —	97 Bk berkelium —	98 Cf californium —	99 Es einsteinium —	100 Fm fermium —	101 Md mendelevium —	102 No nobelium —	103 Lr lawrencium —

The volume of one mole of any gas is 24 dm³ at room temperature and pressure (r.t.p.).



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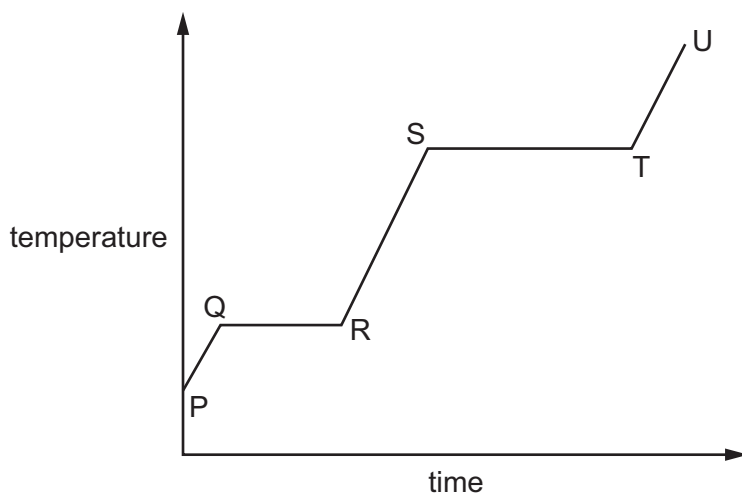
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- 1 What happens to the volume of a fixed mass of gas when the temperature is increased or the pressure is decreased?

	increased temperature	decreased pressure
A	volume decreased	volume decreased
B	volume decreased	volume increased
C	volume increased	volume decreased
D	volume increased	volume increased

- 2 The diagram shows the heating curve for a substance.



Between which two points do the particles of the substance change from being in fixed positions to being mobile?

- A** P and Q **B** Q and R **C** S and T **D** T and U

- 3 A student investigates the diffusion of ammonia gas, NH_3 , and hydrogen chloride gas, HCl .

Two sets of apparatus are set up as shown at room temperature and pressure.



The damp red litmus paper in apparatus 1 changes colour after 30 seconds.

How long does it take for the damp blue litmus paper in apparatus 2 to change colour?

- A 21 seconds
- B 30 seconds
- C 64 seconds
- D The damp blue litmus paper does **not** change colour.

- 4 Which row describes elements, compounds and mixtures?

	elements	compounds	mixtures
A	all the atoms are the same type	contain different atoms chemically joined together	can be separated using physical processes
B	all the atoms are the same type	can be separated into new substances using chemical processes	contain different atoms chemically joined together
C	can be separated into new substances using chemical processes	all the atoms are the same type	contain different atoms chemically joined together
D	contain different atoms chemically joined together	all the atoms are the same type	can be separated using physical processes

- 5 Which statement about atoms or ions is correct?

- A The mass number of an atom is the total number of nucleons and protons.
- B The electronic configuration of an aluminium ion, Al^{3+} , is 2,8,3.
- C Hydrogen is the only atom that has an incomplete first electron shell.
- D Most of the volume of an atom is taken up by its nucleus.

- 6 Which two values are used to calculate the relative atomic mass of a sample of boron?

	value 1	value 2
A	the total number of boron isotopes	the total number of boron atoms
B	the relative abundance of each boron isotope in the sample	the relative mass of each boron isotope
C	the relative abundance of each boron isotope in the sample	the total number of boron atoms
D	the total number of boron isotopes	the relative mass of each boron isotope

- 7 Which statement describes the structure of sodium chloride?

- A** a lattice of alternating positive and negative ions
B a lattice of positive ions surrounded by a 'sea' of delocalised electrons
C a mixture of sodium and chlorine atoms
D a regular lattice of sodium and chlorine atoms

- 8 Graphite and diamond are both forms of the element carbon.

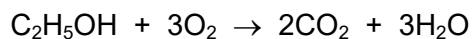
Which row shows the number of other carbon atoms that each carbon atom is covalently bonded to in graphite and in diamond?

	graphite	diamond
A	3	3
B	3	4
C	4	3
D	4	4

- 9 Which row describes the structure and bonding in cobalt?

	structure	bonding
A	giant	ionic
B	simple	ionic
C	giant	metallic
D	simple	metallic

- 10 The equation for the combustion of ethanol is shown.



The relative molecular mass, M_r , of ethanol is 46.

Which statement about the reaction is correct?

- A** 4.6 g of ethanol produces 4.4 g of carbon dioxide.
B 92 g of ethanol produces 88 g of carbon dioxide and 108 g of water.
C 138 g of ethanol reacts with 32 g of oxygen.
D 184 g of ethanol produces 216 g of water.
- 11 A compound contains 40.0% carbon, 6.7% hydrogen and 53.3% oxygen only.

The relative molecular mass, M_r , of the compound is 60.

Which row shows the empirical formula and the molecular formula of the compound?

	empirical formula	molecular formula
A	CH_2O	CH_2O
B	CH_2O	$\text{C}_2\text{H}_4\text{O}_2$
C	$\text{C}_3\text{H}_7\text{O}_3$	$\text{C}_3\text{H}_7\text{O}_3$
D	$\text{C}_3\text{H}_8\text{O}$	$\text{C}_3\text{H}_8\text{O}$

- 12 Hydrogen–oxygen fuel cells are used in electric cars.

Three statements about these fuel cells are listed.

- 1 The process in the cell is called electrolysis.
- 2 Water is broken down in the cell to produce hydrogen, oxygen and electricity.
- 3 Water is the only chemical product formed in the reaction.

Which statements are correct?

- A** 1 and 2 **B** 1 and 3 **C** 2 only **D** 3 only

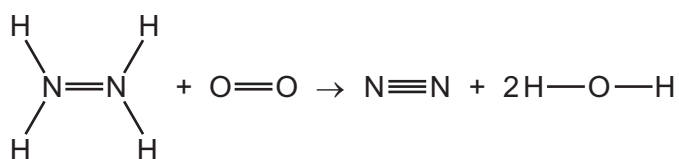
13 A solution of copper(II) sulfate can be electrolysed using copper electrodes or carbon electrodes.

Which statements are correct?

- 1 Using copper electrodes, oxygen gas forms at the anode.
- 2 Using copper electrodes, copper atoms lose electrons at the anode.
- 3 Using carbon electrodes, copper metal forms at the cathode.
- 4 Using carbon electrodes, copper ions gain electrons at the cathode.

A 1 and 2 **B** 1 and 3 **C** 2, 3 and 4 **D** 4 only

14 The equation for the combustion of hydrazine, N_2H_4 , is shown.



bond	bond energy in kJ/mol
N–H	391
N=N	409
N≡N	944
O–H	463
O=O	496

What is the overall enthalpy change for the combustion of one mole of hydrazine?

- A** –327 kJ/mol
B –111 kJ/mol
C +111 kJ/mol
D +327 kJ/mol

- 15** Dissolving ammonium chloride in water is an endothermic change.

Which row shows the energy change and temperature change of the mixture during the dissolving of ammonium chloride?

	energy change	temperature change
A	energy is absorbed	decrease
B	energy is absorbed	increase
C	energy is released	decrease
D	energy is released	increase

- 16** Aqueous sodium thiosulfate, $\text{Na}_2\text{S}_2\text{O}_3$, reacts with dilute hydrochloric acid to form a yellow precipitate of sulfur.



The precipitate forms more quickly when the reactants are heated.

Which statements explain this observation?

- 1 The reacting particles collide more frequently.
- 2 The collisions between reacting particles have more energy.
- 3 The activation energy of the reaction is lower.

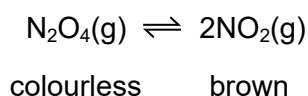
- A** 1 and 2 **B** 1 and 3 **C** 1 only **D** 2 and 3

- 17** Which statement about the conditions used in the Haber process and in the Contact process is correct?

- A** The typical pressure used in the Haber process is similar to that used in the Contact process.
- B** The typical temperature used in the Haber process is similar to that used in the Contact process.
- C** The Haber process uses a vanadium(V) oxide catalyst, and the Contact process uses an iron catalyst.
- D** Both processes produce an acidic gas.

- 18 The equation shows the equilibrium between dinitrogen tetroxide, N_2O_4 , and nitrogen dioxide, NO_2 .

The colours of the reactant and product are shown.



The forward reaction is endothermic.

Which statement is **not** correct?

- A** At equilibrium, the concentrations of the reactant and the product stay constant.
- B** At equilibrium, the rate of the forward reaction is equal to the rate of the reverse reaction.
- C** When the pressure is increased, a darker brown colour is seen.
- D** When the temperature is increased, a darker brown colour is seen.
- 19 What is the oxidation number of chromium in the $\text{Cr}_2\text{O}_7^{2-}$ ion?
- A** -2 **B** -1 **C** +6 **D** +12
- 20 Which method is used to prepare a pure, dry sample of lead(II) chloride?
- A** mix aqueous lead(II) nitrate and dilute hydrochloric acid → filter → crystallise and dry the filtrate
- B** mix aqueous lead(II) nitrate and dilute hydrochloric acid → filter → wash and dry the residue
- C** mix solid lead(II) hydroxide and aqueous sodium chloride → filter → crystallise and dry the filtrate
- D** mix solid lead(II) hydroxide and aqueous sodium chloride → filter → wash and dry the residue
- 21 Phosphorus is an element in Group V of the Periodic Table.

It burns in air to form an oxide that dissolves in water to form a solution with a pH of 1.

Which row describes this oxide of phosphorus?

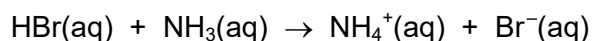
	metal oxide	non-metal oxide	acidic oxide	basic oxide
A	✓	✗	✓	✗
B	✓	✗	✗	✓
C	✗	✓	✓	✗
D	✗	✓	✗	✓

key

✓ = yes

✗ = no

- 22** The equation shows the reaction between aqueous hydrogen bromide and aqueous ammonia.



Which statement describes the role of aqueous hydrogen bromide?

- A** It is a catalyst.
- B** It is a reducing agent.
- C** It is a proton acceptor.
- D** It is a proton donor.

- 23** Which statement about the elements in Group I is correct?

- A** Hydrogen is produced when Group I elements react with water.
- B** Ions of Group I elements have a 1– charge.
- C** Sodium is more reactive than potassium.
- D** Solid sodium is a poor electrical conductor.

- 24** Which row describes the colour and state of iodine at room temperature and pressure?

	colour	state
A	green	liquid
B	grey-black	solid
C	purple	solid
D	red-brown	liquid

- 25** Part of the Periodic Table is shown.

Which element is an inert gas?

A simplified periodic table with 18 columns and 4 rows. The columns are grouped into four sets of three. The elements are labeled A, B, C, and D. A is in the top row, middle column. B is in the second row, right column. C is in the third row, right column. D is in the fourth row, middle column.

26 Four different metals are separately added to water.

The oxide of each metal is separately heated with carbon.

The results are shown.

metal	metal with water	metal oxide with carbon
Q	no reaction	brown solid forms
R	rapid fizzing	no reaction
S	a few bubbles	no reaction
T	a few bubbles	grey solid forms

What is the order of reactivity of the four metals, least reactive first?

A Q → T → S → R

B Q → S → T → R

C R → S → T → Q

D R → T → S → Q

27 An iron nail is fully covered with zinc to prevent it from rusting.

Which row describes this method of protection and explains why zinc is used to protect the iron nail?

	method of protection	reason zinc is used
A	barrier method and sacrificial protection	zinc is more reactive than iron
B	barrier method and sacrificial protection	iron is more reactive than zinc
C	sacrificial protection only	zinc is more reactive than iron
D	barrier method only	iron is more reactive than zinc

28 Iron is extracted from hematite in the blast furnace.

Which reaction increases the temperature in the blast furnace to over 1500 °C?

A calcium carbonate → calcium oxide + carbon dioxide

B calcium oxide + silicon(IV) oxide → calcium silicate

C carbon + oxygen → carbon dioxide

D carbon dioxide + carbon → carbon monoxide

29 Aluminium is extracted by the electrolysis of purified bauxite.

Which row shows the half-equations at the electrodes and explains why one of the electrodes has to be replaced regularly?

	half-equation at anode	half-equation at cathode	electrode replacement
A	$2\text{O}^{2-} \rightarrow \text{O}_2 + 4\text{e}^-$	$\text{Al}^{3+} + 3\text{e}^- \rightarrow \text{Al}$	cathode replaced as it gets coated with aluminium
B	$2\text{O}^{2-} \rightarrow \text{O}_2 + 4\text{e}^-$	$\text{Al}^{3+} + 3\text{e}^- \rightarrow \text{Al}$	anode replaced as it burns away in oxygen
C	$\text{Al}^{3+} + 3\text{e}^- \rightarrow \text{Al}$	$2\text{O}^{2-} \rightarrow \text{O}_2 + 4\text{e}^-$	cathode replaced as it burns away in oxygen
D	$\text{Al}^{3+} + 3\text{e}^- \rightarrow \text{Al}$	$2\text{O}^{2-} \rightarrow \text{O}_2 + 4\text{e}^-$	anode replaced as it gets coated with aluminium

30 Which flow chart describes the correct treatment sequence of the domestic water supply?

- A** filtration → chlorination → sedimentation
- B** sedimentation → chlorination → filtration
- C** chlorination → filtration → sedimentation
- D** sedimentation → filtration → chlorination

31 Which statement about oxides of nitrogen is correct?

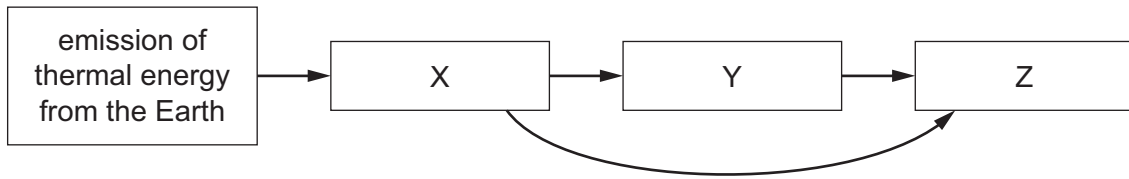
- A** They are made in car engines by the incomplete combustion of gasoline at a high temperature.
- B** They are made in car engines when nitrogen in gasoline reacts with oxygen in air at a high temperature.
- C** They are removed from car exhausts by catalytic converters and form nitrogen and sulfur dioxide.
- D** They are removed from car exhausts when they react with carbon monoxide in catalytic converters.

32 The alkenes are a homologous series.

Which statement about the members of the homologous series of the alkenes is correct?

- A** They have the same functional group.
- B** They have the same relative molecular mass.
- C** They have different chemical properties.
- D** They have different general formulae.

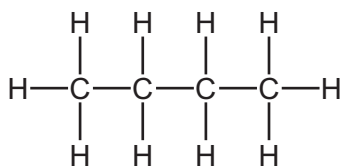
- 33 The diagram represents two processes, X and Y, and their effect, Z, in increasing global warming.



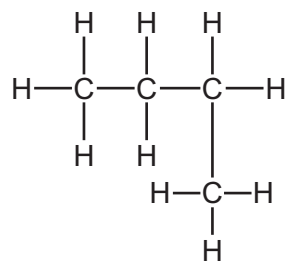
Which row identifies X, Y and Z?

	X	Y	Z
A	absorption of thermal energy by greenhouse gases	emission of thermal energy to the Earth	reduced thermal energy loss to space
B	emission of thermal energy by greenhouse gases	emission of thermal energy to space	reduced thermal energy loss to space
C	absorption of thermal energy by greenhouse gases	emission of thermal energy to the Earth	increased thermal energy loss to space
D	emission of thermal energy by greenhouse gases	emission of thermal energy to space	increased thermal energy loss to space

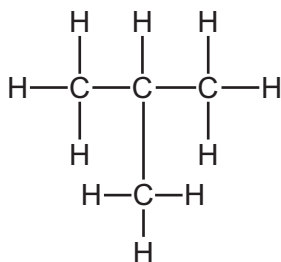
34 Which structures are structural isomers of each other?



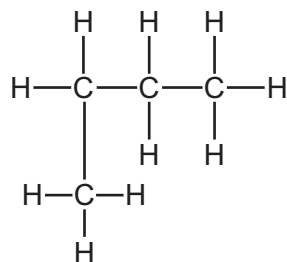
1



2



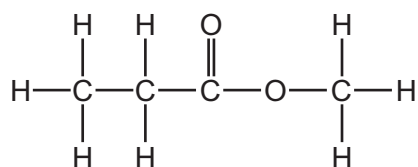
3



4

- A** 1, 2, 3 and 4
B 1, 2 and 4 only
C 1 and 3 only
D 2 and 4 only

35 The structure of an ester is shown.



What is the name of this ester?

- A** ethyl methanoate
B methyl ethanoate
C propyl methanoate
D methyl propanoate

36 Which row shows the products obtained by the cracking of a large alkane?

	alkene	hydrogen	water
A	✓	✓	✓
B	✓	✓	✗
C	✓	✗	✓
D	✗	✓	✓

key

✓ = yes

✗ = no

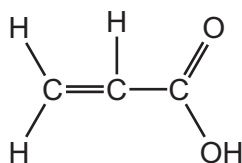
37 Ethanol is produced by the fermentation of aqueous glucose or by the catalytic addition of steam to ethene.

Which row shows some advantages and disadvantages of these processes?

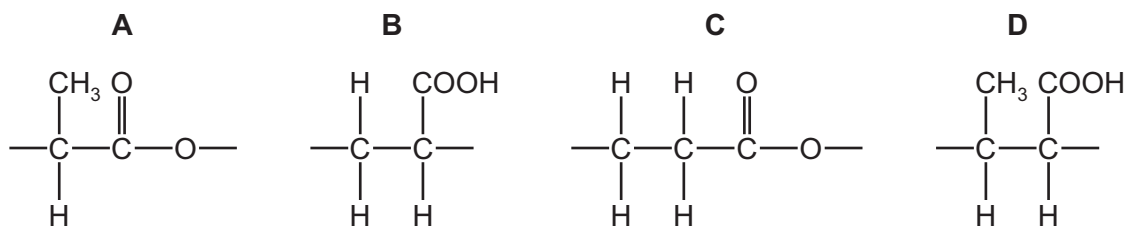
	fermentation of aqueous glucose		catalytic addition of steam to ethene	
	advantage	disadvantage	advantage	disadvantage
A	fast reaction	ethanol produced continuously	almost pure ethanol formed	renewable raw material
B	ethanol produced in batches	slow reaction	fast reaction	ethanol produced continuously
C	renewable raw material	ethanol produced in batches	almost pure ethanol formed	slow reaction
D	renewable raw material	dilute aqueous ethanol formed	fast reaction	non-renewable raw material

38 Propenoic acid is an organic compound that is both an alkene and a carboxylic acid.

The structure of propenoic acid is shown.



Which structure represents one repeat unit of the addition polymer formed from propenoic acid?



39 Which item of apparatus is used to measure 13.7 cm^3 of dilute hydrochloric acid?

- A a balance
- B a burette
- C a conical flask
- D a volumetric pipette

40 Chromatography is used to separate and analyse mixtures of soluble substances.

The first steps of the process are listed.

- apply sample to baseline
- develop chromatogram in solvent

What is the next step in the process to detect soluble **colourless** substances?

- A calculate the R_f values
- B compare the results with known substances
- C count the number of spots
- D use a suitable locating agent

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The Periodic Table of Elements

Group																		
I	II											III	IV	V	VI	VII	VIII	
1Hhydrogen1 Key atomic number atomic symbol name relative atomic mass																		
3Li lithium 7	4Be beryllium 9																	
11Na sodium 23	12Mg magnesium 24																	
13Al aluminium 27	14Si silicon 28	15P phosphorus 31	16S sulfur 32	17Cl chlorine 35.5	18Ar argon 40													
19K potassium 39	20Ca calcium 40	21Sc scandium 45	22Ti titanium 48	23V vanadium 51	24Cr chromium 52	25Mn manganese 55	26Fe iron 56	27Co cobalt 59	28Ni nickel 59	29Cu copper 64	30Zn zinc 65	31Ga gallium 70	32Ge germanium 73	33As arsenic 75	34Se selenium 79	35Br bromine 80	36Kr krypton 84	
37Rb rubidium 85	38Sr strontium 88	39Y yttrium 89	40Zr zirconium 91	41Nb niobium 93	42Mo molybdenum 96	43Tc technetium —	44Ru ruthenium 101	45Rh rhodium 103	46Pd palladium 106	47Ag silver 108	48Cd cadmium 112	49In indium 115	50Sn tin 119	51Sb antimony 122	52Te tellurium 128	53I iodine 127	54Xe xenon 131	
55Cs caesium 133	56Ba barium 137	57–71 lanthanoids		72Hf hafnium 178	73Ta tantalum 181	74W tungsten 184	75Re rhenium 186	76Os osmium 190	77Ir iridium 192	78Pt platinum 195	79Au gold 197	80Hg mercury 201	81Tl thallium 204	82Pb lead 207	83Bi bismuth 209	84Po polonium —	85At astatine —	86Rn radon —
87Fr francium —	88Ra radium —	89–103 actinoids		104Rf rutherfordium —	105Db dubnium —	106Sg seaborgium —	107Bh bohrium —	108Hs hassium —	109Mt meitnerium —	110Ds darmstadtium —	111Rg roentgenium —	112Cn copernicium —	113Nh nihonium —	114Fl flerovium —	115Mc moscovium —	116Lv livermorium —	117Ts tennessine —	118Og oganeson —

lanthanoids	57 La lanthanum 139	58 Ce cerium 140	59 Pr praseodymium 141	60 Nd neodymium 144	61 Pm promethium —	62 Sm samarium 150	63 Eu europium 152	64 Gd gadolinium 157	65 Tb terbium 159	66 Dy dysprosium 163	67 Ho holmium 165	68 Er erbium 167	69 Tm thulium 169	70 Yb ytterbium 173	71 Lu lutetium 175
	89 Ac actinium —	90 Th thorium 232	91 Pa protactinium 231	92 U uranium 238	93 Np neptunium —	94 Pu plutonium —	95 Am americium —	96 Cm curium —	97 Bk berkelium —	98 Cf californium —	99 Es einsteinium —	100 Fm fermium —	101 Md mendelevium —	102 No nobelium —	103 Lr lawrencium —

The volume of one mole of any gas is 24 dm³ at room temperature and pressure (r.t.p.).



Cambridge IGCSE™

CHEMISTRY

0620/23

Paper 2 Multiple Choice (Extended)

October/November 2025

45 minutes

You must answer on the multiple choice answer sheet.

You will need: Multiple choice answer sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)

INSTRUCTIONS

- There are **forty** questions on this paper. Answer **all** questions.
- For each question there are four possible answers **A**, **B**, **C** and **D**. Choose the **one** you consider correct and record your choice in soft pencil on the multiple choice answer sheet.
- Follow the instructions on the multiple choice answer sheet.
- Write in soft pencil.
- Write your name, centre number and candidate number on the multiple choice answer sheet in the spaces provided unless this has been done for you.
- Do **not** use correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.

INFORMATION

- The total mark for this paper is 40.
- Each correct answer will score one mark.
- Any rough working should be done on this question paper.
- The Periodic Table is printed in the question paper.

This document has **16** pages. Any blank pages are indicated.



- 1 What happens to the volume of a fixed mass of gas when pressure and temperature are separately increased?

	volume of gas	
	pressure increased	temperature increased
A	increases	increases
B	no change	decreases
C	decreases	increases
D	no change	increases

- 2 Which gas diffuses most quickly at room temperature and pressure?

A CH₄ **B** C₂H₆ **C** CO **D** SO₂

- 3 Which statement defines nucleon number?

A It is the total number of protons in the nucleus of an atom.
B It is the total number of electrons surrounding the nucleus of an atom.
C It is the total number of neutrons in the nucleus of an atom.
D It is the total number of protons and neutrons in the nucleus of an atom.

- 4 The symbols of four atoms are listed.

1 $^{12}_6\text{C}$

2 $^{14}_6\text{C}$

3 $^{14}_7\text{N}$

4 $^{29}_{14}\text{Si}$

Which atoms have more neutrons than protons?

A 1 and 2 **B** 2 and 3 **C** 2 and 4 **D** 3 and 4

- 5 Which statement describes all positive ions?

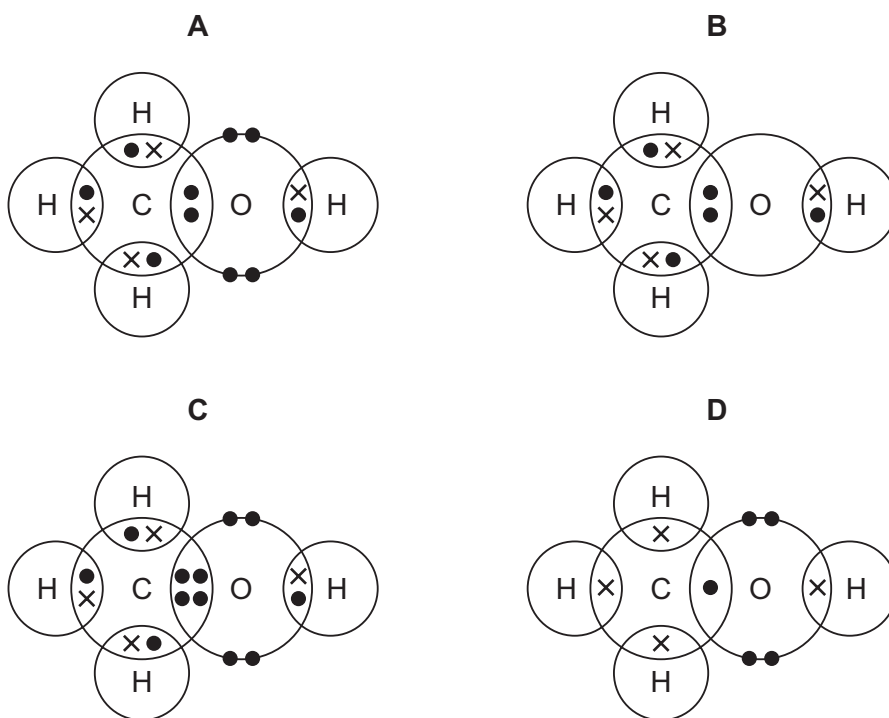
A Positive ions have more electrons than neutrons.
B Positive ions have more protons than neutrons.
C Positive ions have more electrons than protons.
D Positive ions have more protons than electrons.

- 6 A sample of copper has a relative atomic mass of 63.5.

What are the relative abundances of the isotopes in this sample of copper?

- A 25% ^{63}Cu and 75% ^{65}Cu
 B 50% ^{63}Cu and 50% ^{65}Cu
 C 75% ^{63}Cu and 25% ^{65}Cu
 D 90% ^{63}Cu and 10% ^{65}Cu

- 7 Which diagram shows the outer shell electron arrangement for a molecule of methanol, CH_3OH ?



- 8 Rescuers are drilling through fallen rock in order to rescue some people trapped in a cave. The drill needs lubricating from time to time.

Four statements are made about the materials used for the drill tip and the lubricant to explain the reasons for their use.

- 1 Diamond is used for the drill tip as it does **not** conduct electricity.
- 2 Diamond is used for the drill tip as it is very hard.
- 3 Graphite is used as the lubricant as it conducts electricity.
- 4 Graphite is used as the lubricant as it is soft and slippery.

Which statements are correct?

- A 1 and 3 B 1 and 4 C 2 and 3 D 2 and 4

9 Which compound has an empirical formula that is the same as its molecular formula?

- A butane
- B but-1-ene
- C butanoic acid
- D butan-1-ol

10 Which row identifies the charge on the chromium ion and on the sulfate ion in $\text{Cr}_2(\text{SO}_4)_3$?

	Cr	SO_4
A	2+	3–
B	2–	3+
C	3+	2–
D	3–	2+

11 Which statement describes a saturated solution?

- A It is a solution that contains the maximum concentration of solute.
- B It is a solution that contains the maximum amount of solvent.
- C It is a hydrocarbon that contains both double and single bonds.
- D It is a hydrocarbon that contains only single bonds.

12 Calcium carbonate decomposes when heated strongly.



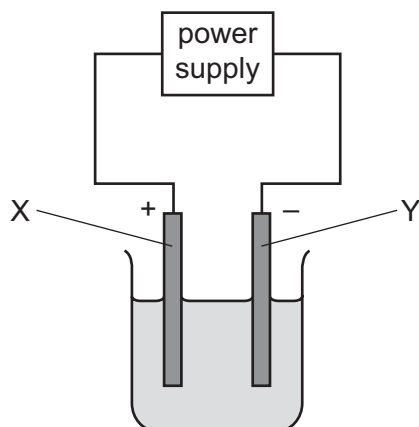
How many gaseous molecules are produced when 10.0 g of calcium carbonate is completely decomposed?

- A 1.00×10^{22} B 1.20×10^{23} C 6.02×10^{22} D 6.02×10^{23}

13 Which product is initially obtained at each electrode during the electrolysis of concentrated aqueous sodium chloride?

	cathode	anode
A	hydrogen	chlorine
B	hydrogen	oxygen
C	sodium	chlorine
D	sodium	oxygen

14 Molten lead(II) bromide is electrolysed using inert electrodes.



Which statement describes the movement of electrons?

- A from X to Y through the external wire
- B from Y to X through the external wire
- C from X to Y through the electrolyte
- D from Y to X through the electrolyte

15 In a fuel cell, a fuel is oxidised to produce electricity.

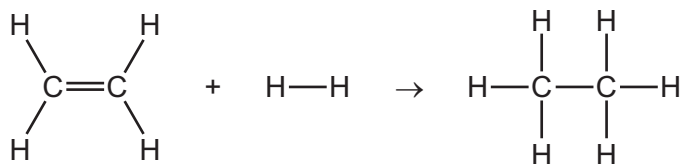
Which row describes the fuel and the oxidising agent used?

	fuel	oxidising agent
A	nitrogen	oxygen
B	hydrogen	oxygen
C	nitrogen	water
D	hydrogen	water

16 Which statement about reactions is correct?

- A A reaction that is vigorous at room temperature has a high activation energy.
- B If ΔH for a reaction is positive, bond making in the formation of the products is endothermic.
- C In an exothermic reaction, the energy required to break bonds is less than the energy released to make bonds.
- D Reactions with a positive enthalpy change always transfer thermal energy to the surroundings.

17 Ethene reacts with hydrogen to form ethane.



The bond energies are shown in the table.

bond	bond energy in kJ/mol
C–C	+350
C–H	+410
H–H	+436
C=C	+614

What is the enthalpy change for the reaction?

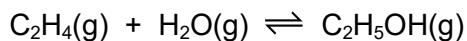
- A –290 kJ/mol
- B –120 kJ/mol
- C +120 kJ/mol
- D +290 kJ/mol

18 Zinc reacts with dilute hydrochloric acid.

Which statement about the particles in the reaction is correct?

- A Increasing the concentration of dilute hydrochloric acid increases the collision rate but has **no** effect on the activation energy.
- B Increasing the concentration of dilute hydrochloric acid increases the collision rate and the activation energy.
- C Increasing the temperature of the reaction increases the activation energy.
- D Increasing the temperature of the reaction causes all collisions to lead to a reaction.

- 19 Ethanol is manufactured by the addition of steam to ethene.



The reaction conditions are 300 °C and 60 atm.

The forward reaction is exothermic.

Which rows describe the effect on the equilibrium yield of C₂H₅OH when the stated change is made?

	change	equilibrium yield of C ₂ H ₅ OH
1	adding a catalyst	no change
2	decreasing the temperature	increases
3	decreasing the pressure	increases

- A** 1 and 2 **B** 1 and 3 **C** 2 only **D** 3 only

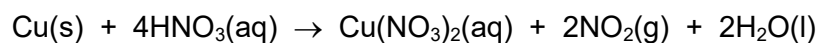
- 20 Which row identifies the conditions used to make sulfur trioxide in the Contact process?

	pressure / kPa	catalyst
A	200	iron
B	200	vanadium(V) oxide
C	20 000	iron
D	20 000	vanadium(V) oxide

- 21 Which statement describes a base?

- A** It reacts with ammonium chloride to produce ammonia gas.
B It reacts with calcium carbonate to produce carbon dioxide gas.
C It reacts with copper to produce hydrogen gas.
D It turns blue litmus red.

- 22** The equation shows the reaction between copper and concentrated nitric acid.



Which element is reduced in this reaction?

- A** oxygen
B nitrogen
C copper
D hydrogen

- 23** The Periodic Table shows the position of four different elements.

A simplified periodic table with 18 columns and 4 rows. The first two columns are labeled A and C. The next 10 columns are empty. The last two columns are labeled B and D. A small square box is located above the 10th column.

Which row is correct?

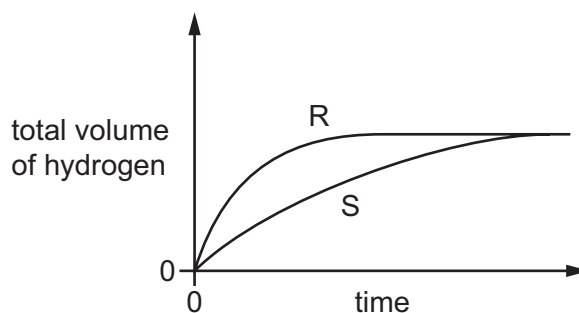
	element	number of electrons in the outer shell
A	non-metal	1
B	metal	5
C	metal	2
D	non-metal	7

24 Dilute acid R and dilute acid S have the same concentration.

The same volume of each dilute acid is reacted separately with the same mass of magnesium ribbon. Both reactions are carried out at room temperature and pressure.

The volume of hydrogen produced is measured.

A graph of the results is shown.



Which statement about the reactions is correct?

- A** Acid S reacts faster than acid R.
- B** The final volume of hydrogen produced in each reaction is different.
- C** Acid R is a stronger acid than acid S.
- D** Acid S is a stronger acid than acid R.

25 Which metal oxide reacts with aqueous sodium hydroxide to form a salt and water?

- A** calcium oxide
- B** copper(II) oxide
- C** sodium oxide
- D** aluminium oxide

26 Which row identifies a transition element?

	melting point	density at r.t.p.
A	high	high
B	high	low
C	low	high
D	low	low

27 Which statement about Group I or Group VII elements is correct?

- A Group VII elements are monatomic non-metals.
- B Lithium is more reactive with water than caesium.
- C The melting point of Group I metals increases down the group.
- D Potassium bromide reacts with chlorine to produce an orange solution.

28 Some information about four metals P, Q, R and S is listed.

Metal P does **not** react with dilute hydrochloric acid.

Metal Q is more reactive than magnesium.

Metal R reacts rapidly with cold water to produce hydrogen.

Metal S reacts with dilute hydrochloric acid to produce hydrogen.

Which row identifies P, Q, R and S?

	P	Q	R	S
A	gold	aluminium	sodium	copper
B	copper	potassium	calcium	magnesium
C	silver	calcium	iron	zinc
D	magnesium	sodium	potassium	iron

29 Cars are fitted with catalytic converters.

Which equation represents a reaction that takes place in a catalytic converter?

- A $\text{CO}_2 + \text{NO}_2 \rightarrow \text{CO} + \text{NO} + \text{O}_2$
- B $2\text{CO} + 2\text{NO} \rightarrow 2\text{CO}_2 + \text{N}_2$
- C $\text{CO}_2 + \text{N}_2 \rightarrow \text{C} + 2\text{NO}$
- D $\text{C} + \text{NO}_2 \rightarrow \text{CO} + \text{NO}$

30 Three statements about greenhouse gases and global warming are listed.

- 1 A greenhouse gas molecule can absorb and emit thermal energy.
- 2 The Earth is a source of thermal energy.
- 3 Thermal energy emitted by a greenhouse gas molecule is sent out in all directions.

Which statements are correct?

- A 1, 2 and 3
- B 1 and 2 only
- C 1 and 3 only
- D 2 and 3 only

31 Which type of reaction takes place when methane reacts with chlorine in the presence of ultraviolet light?

- A addition
- B cracking
- C polymerisation
- D substitution

32 Iron is extracted from iron(III) oxide in the blast furnace.

Three steps in the process are listed.

- 1 Carbon (coke) is completely oxidised.
- 2 Iron(III) oxide is reduced.
- 3 Carbon dioxide is reduced to carbon monoxide.

What is the correct order of the steps in the extraction of iron?

- A 1 → 2 → 3 B 1 → 3 → 2 C 2 → 1 → 3 D 3 → 1 → 2

33 Three different reactions that occur during the extraction of aluminium from purified bauxite are listed.

- 1 $\text{Al}^{3+} + 3\text{e}^- \rightarrow \text{Al}$
- 2 $\text{C} + \text{O}_2 \rightarrow \text{CO}_2$
- 3 $2\text{O}^{2-} \rightarrow \text{O}_2 + 4\text{e}^-$

Which reactions occur at the anode?

- A 1 and 2 B 1 only C 2 and 3 D 3 only

34 Which statement about aqueous ethanoic acid is correct?

- A It reacts with magnesium to form oxygen gas.
- B It reacts with sodium carbonate to form carbon dioxide gas.
- C It turns red litmus paper blue.
- D It turns methyl orange yellow.

35 The diagrams represent four monomers.



The monomers each react in a 1 : 1 ratio with the dicarboxylic acid shown.

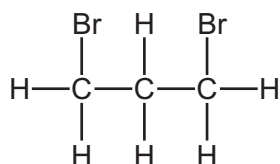


How many of the four monomers shown can react with the dicarboxylic acid to form a polymer?

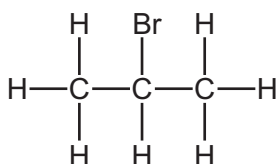
- A** 1 **B** 2 **C** 3 **D** 4

36 Propene undergoes addition reactions with bromine, hydrogen and steam.

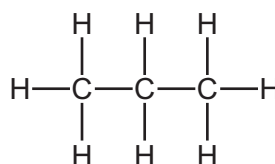
Which structures represent a product of these reactions?



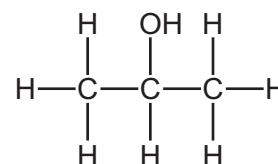
1



2



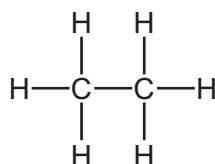
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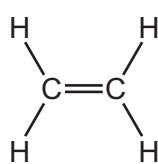
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- A** 1 and 3 **B** 1 and 4 **C** 2 and 3 **D** 3 and 4

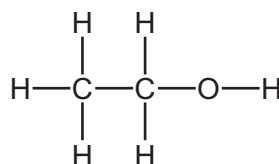
37 The structures of four compounds are shown.



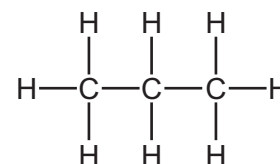
1



2



3



4

Which compounds are members of the same homologous series?

- A** 1, 2 and 3 **B** 1 and 2 only **C** 1 and 4 **D** 2 and 4

38 Which statement about ethanoic acid is correct?

- A** Ethanol is reduced by bacteria to produce ethanoic acid.
B Ethanoic acid reacts with magnesium to produce a salt of formula CH_3COOMg .
C Ethanoic acid and methanol react in the presence of an acid catalyst to produce ethyl methanoate.
D Ethanol reacts with acidified aqueous potassium manganate(VII) to produce ethanoic acid.

39 Substance X contains one cation.

Two tests are completed on aqueous X.

test 1 flame test

test 2 addition of aqueous sodium hydroxide dropwise and then in excess

Which row shows a possible combination of observations for the tests?

	observation for test 1	observation for test 2
A	yellow flame	white ppt., insoluble in excess
B	blue-green flame	light blue ppt., insoluble in excess
C	light green flame	green ppt., soluble in excess
D	red flame	red-brown ppt., insoluble in excess

40 A colourless mixture is separated by paper chromatography.

The steps below describe the method. These steps are **not** in the correct order.

- 1 Remove the chromatography paper from the solvent when the solvent front is near the top of the paper.
- 2 Place the chromatography paper in the solvent.
- 3 Spray the chromatography paper with a locating agent.
- 4 Place a spot of the mixture on the pencil line.
- 5 Draw a pencil line at the bottom of the chromatography paper.
- 6 Draw a pencil line to mark the position of the solvent front.

What is the correct order for these steps?

- A** 3 → 5 → 6 → 4 → 2 → 1
- B** 5 → 4 → 3 → 2 → 1 → 6
- C** 3 → 5 → 4 → 1 → 2 → 6
- D** 5 → 4 → 2 → 1 → 6 → 3

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The Periodic Table of Elements

Group																		
I	II											III	IV	V	VI	VII	VIII	
3 Li lithium 7	4 Be beryllium 9	Key atomic number atomic symbol name relative atomic mass										1 H hydrogen 1	5 B boron 11	6 C carbon 12	7 N nitrogen 14	8 O oxygen 16	9 F fluorine 19	10 Ne neon 20
11 Na sodium 23	12 Mg magnesium 24											13 Al aluminium 27	14 Si silicon 28	15 P phosphorus 31	16 S sulfur 32	17 Cl chlorine 35.5	18 Ar argon 40	
19 K potassium 39	20 Ca calcium 40	21 Sc scandium 45	22 Ti titanium 48	23 V vanadium 51	24 Cr chromium 52	25 Mn manganese 55	26 Fe iron 56	27 Co cobalt 59	28 Ni nickel 59	29 Cu copper 64	30 Zn zinc 65	31 Ga gallium 70	32 Ge germanium 73	33 As arsenic 75	34 Se selenium 79	35 Br bromine 80	36 Kr krypton 84	
37 Rb rubidium 85	38 Sr strontium 88	39 Y yttrium 89	40 Zr zirconium 91	41 Nb niobium 93	42 Mo molybdenum 96	43 Tc technetium —	44 Ru ruthenium 101	45 Rh rhodium 103	46 Pd palladium 106	47 Ag silver 108	48 Cd cadmium 112	49 In indium 115	50 Sn tin 119	51 Sb antimony 122	52 Te tellurium 128	53 I iodine 127	54 Xe xenon 131	
55 Cs caesium 133	56 Ba barium 137	57–71 lanthanoids	72 Hf hafnium 178	73 Ta tantalum 181	74 W tungsten 184	75 Re rhenium 186	76 Os osmium 190	77 Ir iridium 192	78 Pt platinum 195	79 Au gold 197	80 Hg mercury 201	81 Tl thallium 204	82 Pb lead 207	83 Bi bismuth 209	84 Po polonium —	85 At astatine —	86 Rn radon —	
87 Fr francium —	88 Ra radium —	89–103 actinoids	104 Rf rutherfordium —	105 Db dubnium —	106 Sg seaborgium —	107 Bh bohrium —	108 Hs hassium —	109 Mt meitnerium —	110 Ds darmstadtium —	111 Rg roentgenium —	112 Cn copernicium —	113 Nh nihonium —	114 Fl flerovium —	115 Mc moscovium —	116 Lv livermorium —	117 Ts tennessine —	118 Og oganeson —	

lanthanoids		57 La lanthanum 139	58 Ce cerium 140	59 Pr praseodymium 141	60 Nd neodymium 144	61 Pm promethium —	62 Sm samarium 150	63 Eu europium 152	64 Gd gadolinium 157	65 Tb terbium 159	66 Dy dysprosium 163	67 Ho holmium 165	68 Er erbium 167	69 Tm thulium 169	70 Yb ytterbium 173	71 Lu lutetium 175
		89 Ac actinium —	90 Th thorium 232	91 Pa protactinium 231	92 U uranium 238	93 Np neptunium —	94 Pu plutonium —	95 Am americium —	96 Cm curium —	97 Bk berkelium —	98 Cf californium —	99 Es einsteinium —	100 Fm fermium —	101 Md mendelevium —	102 No nobelium —	103 Lr lawrencium —

The volume of one mole of any gas is 24 dm³ at room temperature and pressure (r.t.p.).

Cambridge IGCSE™

CHEMISTRY**0620/22**

Paper 2 Multiple Choice (Extended)

February/March 2026**45 minutes**

You must answer on the multiple choice answer sheet.

You will need: Multiple choice answer sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)

INSTRUCTIONS

- There are **forty** questions on this paper. Answer **all** questions.
- For each question there are four possible answers **A, B, C** and **D**. Choose the **one** you consider correct and record your choice in soft pencil on the multiple choice answer sheet.
- Follow the instructions on the multiple choice answer sheet.
- Write in soft pencil.
- Write your name, centre number and candidate number on the multiple choice answer sheet in the spaces provided unless this has been done for you.
- Do **not** use correction fluid or tape.
- Do **not** write on any bar codes.
- You may use a calculator.

INFORMATION

- The total mark for this paper is 40.
- Each correct answer will score one mark.
- Any rough working should be done on this question paper.
- The Periodic Table is printed in the question paper.

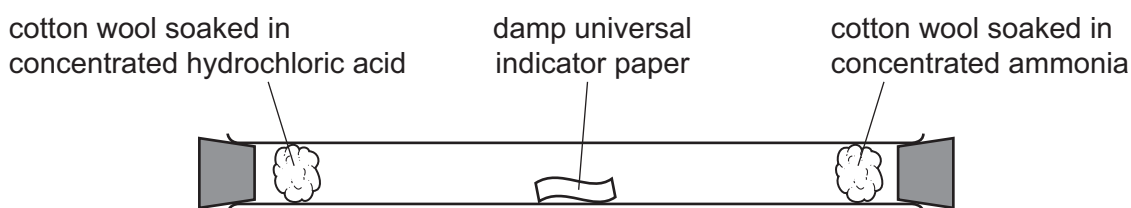
This document has **16** pages. Any blank pages are indicated.

- 1 A sample of helium gas is placed in a free-moving gas syringe.

The temperature of the helium is increased. The pressure of the helium does **not** change.

Which statement is correct?

- A** The volume of helium decreases, and the helium atoms get closer together.
B The volume of helium decreases, and the helium atoms get further apart.
C The volume of helium increases, and the helium atoms get closer together.
D The volume of helium increases, and the helium atoms get further apart.
- 2 An experiment is set up as shown.



The concentrated hydrochloric acid gives off hydrogen chloride gas, HCl , which dissolves in water to form an acid.

The concentrated ammonia gives off ammonia gas which dissolves in water to form an alkali.

Which row describes and explains the first change in the universal indicator paper?

	colour of universal indicator paper	reason
A	blue	molecules of ammonia move more quickly than molecules of hydrogen chloride
B	red	molecules of ammonia move more quickly than molecules of hydrogen chloride
C	blue	molecules of hydrogen chloride move more quickly than molecules of ammonia
D	red	molecules of hydrogen chloride move more quickly than molecules of ammonia

- 3** X^{2+} is an ion of a Group 2 element, X.

What has the highest value in the ion?

- A** the nucleon number
B the number of electrons
C the number of neutrons
D the proton number

- 4 The positions of two elements, P and Q, in the Periodic Table are shown.

A 10x4 grid representing a periodic table. The first two columns are on the left, and the last two are on the right, with a gap in the middle. The grid is divided into four rows. The first row has a single box in the 10th column. The second row has boxes in columns 1, 2, 8, 9, and 10. The third row has boxes in columns 1, 2, 3, 4, 5, 6, 7, 8, 9, and 10. The fourth row has boxes in columns 1, 2, 3, 4, 5, 6, 7, 8, 9, and 10. The letter 'Q' is in the first column, third row. The letter 'P' is in the 9th column, third row.

P and Q react together to form a compound.

What is the formula of the compound?

- A** QP **B** Q₂P **C** Q₇P **D** QP₇

- 5** Which row describes the properties of a simple molecular compound?

	melting point /°C	boiling point /°C	electrical conductivity at room temperature and pressure
A	−39	357	good
B	3550	4830	poor
C	11	181	poor
D	97	882	good

- 6** How are the structures of diamond and silicon(IV) oxide similar?

- A** Molecules of both diamond and silicon(IV) oxide are held together by weak attractive forces.
- B** They both contain atoms arranged in planes held together by weak bonds.
- C** They both contain ions that are free to move.
- D** The carbon in diamond and the silicon in silicon(IV) oxide each have four covalent bonds.

- 7 Solid X and solid Y are both pure substances that contain ions.

X is **not** an electrical conductor.

Y is an electrical conductor.

Which statements about X and Y are correct?

- 1 X is a compound. Y is an element.
- 2 The ions in X are held in a lattice. The ions in Y are **not** held in a lattice.
- 3 X contains both positive and negative ions. Y contains positive ions but no negative ions.

A 1, 2 and 3 **B** 1 and 2 only **C** 1 and 3 only **D** 2 and 3 only

- 8 A sample of magnesium is burned in air.

A student calculates that 10 g of magnesium oxide should be produced.

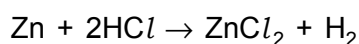
After the reaction, the student determines the percentage yield of magnesium oxide.

Which formula is used to determine the percentage yield of magnesium oxide?

- A** $\frac{\text{relative atomic mass of magnesium}}{\text{relative formula mass of magnesium oxide}} \times 100$
- B** $\frac{\text{mass of magnesium oxide formed}}{\text{mass of magnesium used}} \times 100$
- C** $\frac{\text{mass of magnesium oxide formed}}{\text{mass of magnesium oxide expected}} \times 100$
- D** $\frac{\text{mass of pure magnesium oxide}}{\text{mass of all reaction products}} \times 100$

- 9 4.55 g of zinc is reacted with 50 cm³ of 2.25 mol/dm³ dilute hydrochloric acid.

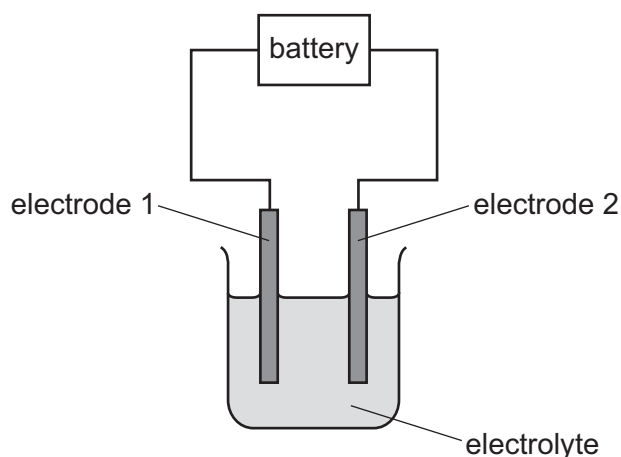
The equation for the reaction is shown.



Which volume of hydrogen gas, at room temperature and pressure, is produced in the reaction?

- A** 1.35 dm³ **B** 1.68 dm³ **C** 2.70 dm³ **D** 3.36 dm³

- 10 In the electrolysis diagram, oxidation is occurring at electrode 1 and reduction at electrode 2.



Which row shows the directions of movement of the electrons in the external circuit and of the positive ions in the electrolyte?

	direction of movement of electrons in external circuit	direction of movement of positive ions in electrolyte
A	1 → 2	1 → 2
B	1 → 2	2 → 1
C	2 → 1	1 → 2
D	2 → 1	2 → 1

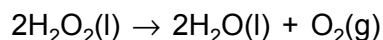
- 11 Which row describes an advantage of using the hydrogen–oxygen fuel cell and an advantage of using the petrol engine?

	advantage of using the hydrogen–oxygen fuel cell	advantage of using the petrol engine
A	can travel a greater distance after the fuel tank has been filled	does not produce carbon dioxide gas which causes global warming
B	there are more petrol filling stations than hydrogen filling stations	no risk of fire or explosion if the fuel tank is damaged
C	does not produce carbon dioxide gas which causes global warming	can travel a greater distance after the fuel tank has been filled
D	no risk of fire or explosion if the fuel tank is damaged	there are more petrol filling stations than hydrogen filling stations

12 Which statement describes activation energy?

- A It is the maximum energy given out during a reaction.
- B It is the maximum energy taken in during a reaction.
- C It is the minimum energy colliding particles must have to react.
- D It is the minimum energy particles must have to make bonds.

13 Hydrogen peroxide, H–O–O–H, decomposes to form water and oxygen.



The bond energies are shown in the table. The reaction is exothermic.

bond	bond energy in kJ/mol
O–H	+460
O–O	+150
O=O	+496

What is the enthalpy change for this reaction?

- A –346 kJ/mol B –196 kJ/mol C +196 kJ/mol D +346 kJ/mol

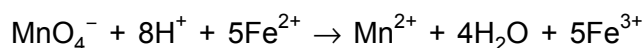
14 Which row shows the sources of nitrogen and of hydrogen for the Haber process?

	source of nitrogen	source of hydrogen
A	nitrates	water
B	nitrates	methane
C	air	water
D	air	methane

15 Which statement about anhydrous salts is correct?

- A Anhydrous cobalt(II) chloride turns from white to blue when added to water.
- B Anhydrous cobalt(II) chloride loses water on gentle heating.
- C Anhydrous copper(II) sulfate reacts with water to make hydrated copper(II) sulfate.
- D Anhydrous copper(II) sulfate turns from blue to white when water is added.

- 16 The equation represents the reaction of iron(II) sulfate with acidified aqueous potassium manganate(VII).



Which ion is reduced and what is the reducing agent?

	ion reduced	reducing agent
A	Fe^{2+}	Fe^{2+}
B	Fe^{2+}	MnO_4^-
C	MnO_4^-	Fe^{2+}
D	MnO_4^-	MnO_4^-

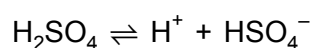
- 17 Which statements about aqueous hydrochloric acid are correct?

- 1 It is completely dissociated in aqueous solution.
- 2 It turns thymolphthalein blue.
- 3 It acts as a strong acid in aqueous solution.
- 4 It produces OH^- ions.

- A** 1 and 3 **B** 1 and 4 **C** 2 and 3 **D** 2 and 4

- 18 Sulfuric acid dissociates in water in two stages.

The equations for each stage are shown.



Which statements about these equations are correct?

- 1 HSO_4^- can donate and accept protons.
- 2 H_2SO_4 is a proton donor.
- 3 SO_4^{2-} is the only base.

- A** 1 and 2 **B** 1 only **C** 2 and 3 **D** 3 only

19 Which statement about amphoteric oxides is correct?

- A They react with acids but **not** bases to form a salt and water.
- B They react with acids and bases to form a salt and water.
- C They react with acids and bases to form a salt and hydrogen.
- D They react with bases but **not** acids to form a salt and water.

20 Which property of elements increases across a period of the Periodic Table?

- A metallic character
- B number of electron shells
- C number of outer-shell electrons
- D tendency to form positive ions

21 Which two compounds react together to form the insoluble salt lead(II) chloride?

	compound	solubility in water
1	lead(II) nitrate	yes
2	lead(II) sulfate	no
3	silver chloride	no
4	sodium chloride	yes

- A 1 and 3 B 1 and 4 C 2 and 3 D 2 and 4

22 Which statements describe properties of transition elements?

- 1 They form white compounds only.
- 2 They are ductile.
- 3 They have high densities.
- 4 They form positive ions by gaining electrons.

- A 1 and 2 B 1 and 4 C 2 and 3 D 3 and 4

23 Which statement explains why aluminium has good resistance to corrosion?

- A** Aluminium is a highly reactive metal.
- B** Aluminium has an oxide layer on its surface.
- C** Aluminium requires heat to react.
- D** Aluminium is an inert metal.

24 Coating iron helps to prevent rusting.

Which coating will continue to protect the iron even when the coating is damaged?

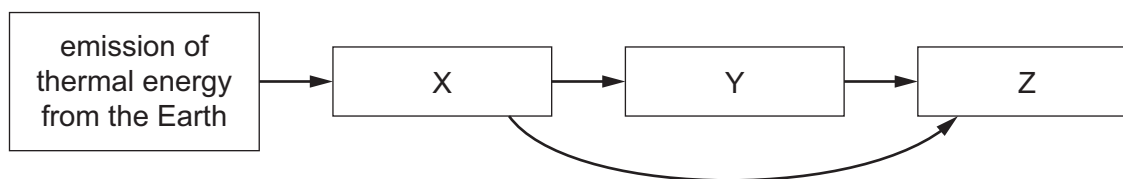
- A** copper
- B** paint
- C** plastic
- D** zinc

25 Which statements about the substances in water from natural sources are correct?

- 1 Detergents contain nitrates that cause disease.
- 2 Sewage contains harmful microbes that cause disease.
- 3 All metal compounds are toxic.
- 4 Fertilisers contain nitrates and phosphates which damage aquatic life.

- A** 1 and 3 **B** 1 and 4 **C** 2 and 3 **D** 2 and 4

- 26 The diagram represents processes X and Y and their effect, Z, in increasing global warming.



Which row identifies X, Y and Z?

	X	Y	Z
A	absorption of thermal energy by greenhouse gases	emission of thermal energy to the Earth	reduced thermal energy loss to space
B	emission of thermal energy by greenhouse gases	emission of thermal energy to space	reduced thermal energy loss to space
C	absorption of thermal energy by greenhouse gases	emission of thermal energy to the Earth	increased thermal energy loss to space
D	emission of thermal energy by greenhouse gases	emission of thermal energy to space	increased thermal energy loss to space

- 27 Which equation describes the formation of two atmospheric pollutants that are **not** present in clean air?

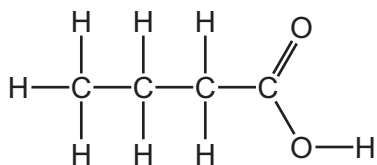
- A $2\text{C}_6\text{H}_{14} + 19\text{O}_2 \rightarrow 12\text{CO}_2 + 14\text{H}_2\text{O}$
- B $2\text{C}_7\text{H}_{14} + 18\text{O}_2 \rightarrow 8\text{CO}_2 + 6\text{CO} + 14\text{H}_2\text{O}$
- C $2\text{C}_8\text{H}_{18} + 19\text{O}_2 \rightarrow 10\text{CO}_2 + 6\text{C} + 18\text{H}_2\text{O}$
- D $2\text{C}_9\text{H}_{18} + 16\text{O}_2 \rightarrow 14\text{CO} + 4\text{C} + 18\text{H}_2\text{O}$

- 28 Some of the fractions obtained from the fractional distillation of petroleum are used as fuels for vehicles.

Which two fractions are used as fuels for vehicles?

- A bitumen fraction and gasoline fraction
- B bitumen fraction and naphtha fraction
- C gasoline fraction and kerosene fraction
- D kerosene fraction and lubricating fraction

29 The displayed formula of butanoic acid is shown.



Butanoic acid reacts with ethanol to form an ester.

What is the structural formula and the empirical formula of the ester?

	structural formula	empirical formula
A	$\text{C}_3\text{H}_7\text{COOC}_2\text{H}_5$	$\text{C}_6\text{H}_{12}\text{O}_2$
B	$\text{C}_3\text{H}_7\text{COOC}_2\text{H}_5$	$\text{C}_3\text{H}_6\text{O}$
C	$\text{CH}_3\text{COOC}_4\text{H}_9$	$\text{C}_6\text{H}_{12}\text{O}_2$
D	$\text{CH}_3\text{COOC}_4\text{H}_9$	$\text{C}_3\text{H}_6\text{O}$

30 What is the structural formula of propan-1-ol?

- A** $\text{C}_3\text{H}_7\text{OH}$
- B** $\text{CH}_3\text{CH}_2\text{CH}_2\text{OH}$
- C** $\text{C}_3\text{H}_8\text{O}$
- D** $\text{C}_n\text{H}_{2n+1}\text{OH}$

31 Alkenes can undergo addition reactions with aqueous bromine, hydrogen or steam.

Which reactions of ethene need a catalyst?

- 1 with aqueous bromine
 - 2 with steam
 - 3 with hydrogen
- A** 1, 2 and 3 **B** 1 and 2 only **C** 1 only **D** 2 and 3 only

32 Which conditions are used to manufacture ethanol by fermentation?

	temperature / °C	catalyst
A	300	yeast
B	30	acid
C	300	acid
D	30	yeast

33 Which statement about an aqueous solution of ethanoic acid is correct?

- A** It reacts with magnesium to form water as one of the products.
- B** It reacts with sodium carbonate to form carbon dioxide.
- C** It reacts with sodium hydroxide to form hydrogen.
- D** It turns red litmus paper blue.

34 Which statements explain why plastics should be recycled?

- 1 Most do **not** decompose when they are added to landfill.
- 2 They pollute rivers and the ocean where they harm wildlife.
- 3 They can produce toxic gases when they are burned.

A 1, 2 and 3 **B** 1 and 2 only **C** 1 and 3 only **D** 2 and 3 only

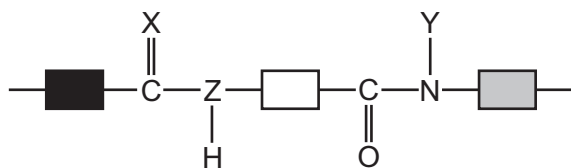
35 Which statements about the plastic PET are correct?

- 1 PET is a condensation polymer.
- 2 PET is a polyamide.
- 3 PET can be converted into monomers and re-polymerised.

A 1, 2 and 3 **B** 1 and 2 only **C** 1 and 3 only **D** 2 and 3 only

36 The diagram represents part of the structure of a protein.

Three atoms have been labelled X, Y and Z.



Which elements are X, Y and Z?

	X	Y	Z
A	carbon	hydrogen	oxygen
B	hydrogen	oxygen	nitrogen
C	oxygen	hydrogen	nitrogen
D	oxygen	hydrogen	carbon

37 Locating agents are often used in paper chromatography.

Which mixture requires a locating agent?

- A a mixture of two colourless substances
- B a mixture of two insoluble substances
- C a mixture of two different-coloured substances with the same R_f value
- D a mixture of two substances that are the same colour

38 A blue solid, Q, is soluble in water.

Which method is used to obtain pure solid Q from aqueous solution?

- A chromatography
- B crystallisation
- C filtration
- D neutralisation

- 39** Two test-tubes each contain a sample of water. One test-tube contains pure water, but the other test-tube contains water and dissolved sodium chloride as an impurity.

Which tests may be used to identify the test-tube that contains pure water?

- 1 Measure the boiling point of each water sample.
- 2 Test each sample with acidified aqueous silver nitrate.
- 3 Test each sample with anhydrous copper(II) sulfate.

A 1 and 2 **B** 1 only **C** 2 and 3 **D** 3 only

- 40** The results of some tests on dilute aqueous solution X are shown.

test	observation
add aqueous ammonia dropwise	white precipitate
then add aqueous ammonia in excess	precipitate dissolves to form colourless solution
add aqueous sodium hydroxide dropwise	white precipitate
then add aqueous sodium hydroxide in excess	precipitate dissolves to form colourless solution

Which cation is present in X?

A Al^{3+} **B** Ca^{2+} **C** Cr^{3+} **D** Zn^{2+}

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The Periodic Table of Elements

Group															
I	II	Key						III	IV	V	VI	VII	VIII		
		atomic number atomic symbol name relative atomic mass						1 H hydrogen 1							
3 Li lithium 7	4 Be beryllium 9														2 He helium 4
11 Na sodium 23	12 Mg magnesium 24														9 F fluorine 19
															8 O oxygen 16
															7 N nitrogen 14
															6 C carbon 12
															5 B boron 11
															4 Ne neon 20
															3 Cl chlorine 35.5
															2 Ar argon 40
															36 Kr krypton 84
															54 Xe xenon 131
															86 Rn radon —
															116 Lv livermorium —
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lanthanoids

57 La lanthanum 139	58 Ce cerium 140	59 Pr praseodymium 141	60 Nd neodymium 144	61 Pm promethium —	62 Sm samarium 150	63 Eu europium 152	64 Gd gadolinium 157	65 Tb terbium 159	66 Dy dysprosium 163	67 Ho holmium 165	68 Er erbium 167	69 Tm thulium 169	70 Yb ytterbium 173	71 Lu lutetium 175
89 Ac actinium —	90 Th thorium 232	91 Pa protactinium 231	92 U uranium 238	93 Np neptunium —	94 Pu plutonium —	95 Am americium —	96 Cm curium —	97 Bk berkelium —	98 Cf californium —	99 Es einsteinium —	100 Fm fermium —	101 Md mendelevium —	102 No nobelium —	103 Lr lawrencium —

actinoids

The volume of one mole of any gas is 24 dm³ at room temperature and pressure (r.t.p.)